

Brinda Udayakumar

Ph. No +1 (408) 304-8018 | Email: budayaku@asu.edu | Tempe, AZ

ACADEMIC QUALIFICATION

Master of Science (Computer Science)

Aug 2025 – May 2027

Arizona State University, Tempe, AZ | GPA: 3.93/4

[Coursework: Advanced Operating Systems, Distributed Database Systems, Natural Language Processing, Foundations of Algorithms, Image Processing and Analysis, Knowledge Representation and Reasoning, Artificial Intelligence, Statistical Machine Learning]

Bachelor of Engineering (Computer Science and Engineering)

Aug 2017 – Jul 2021

SRM Easwari Engineering College, Chennai, India | GPA: 9.36/10

EXPERIENCE

Graduate Research Assistant, Arizona State University, Tempe, AZ

Mar 2026 – Present

Project: Autonomous Multi-Agent Counter-UAS System

- Designed a real-time multi-agent task allocation algorithm assigning defending UAVs to incoming targets under a strict no-overlap constraint, using a predictive time-to-intercept cost model with hysteresis-based switching to eliminate reassignment instability.
- Built a Python simulation framework evaluating performance across thousands of randomized engagements, isolating sensing range as the dominant limiting factor through controlled ablation across force ratio, platform speed, and sensor range.
- Trained and evaluated YOLO11-OB models for drone detection on RGB and synthetic infrared imagery using an NVIDIA A100 HPC cluster, building XAI and non-XAI variants to compare detection performance against explainability.
- Developed a custom OBB evaluation pipeline using Shapely-based rotated IoU to compute per-class precision, recall, and mAP.

Software Engineer, LTIMindtree, Chennai, India

Aug 2021 – Sep 2024

- Proactively designed and implemented a Python automation pipeline that replaced a highly manual, multi person workflow, compressing ~6 months of manual effort into about one week of automated processing and earning formal appreciation from management.
- Owned the design and development of a new functional module for an existing application, translating business requirements into robust data models, business rules, and user journeys with minimal supervision.
- Built and enhanced end to end features on the Jitterbit low code platform, developing front end flows, configuring middle tier logic, and integrating with MySQL Server back ends for HR, payroll, and requisition systems used by multiple enterprise clients.
- Provided production support for multiple clients: triaged incident tickets, performed root cause analysis across Jitterbit configurations and SQL queries, and shipped fixes that resolved incidents and reduced recurrence in live environments.
- Delivered both major and minor product enhancements across a suite of 5 production applications supporting core business operations, driving improvements in usability, performance, and reliability from requirements clarification through deployment.
- Collaborated with a ~15 member cross functional team (developers, QA, product owners, and client stakeholders) to prioritize work, estimate effort, and deliver features and fixes on schedule while maintaining application stability and data accuracy.

Software Engineer, New England Software Technology, Texas, United States (Freelance)

May 2024 – Jul 2024

- Architected and launched the company's full-stack website.
- Engineered a key product integration feature using Python-Flask, creating a direct sales and information portal for the company's flagship offering.
- Delivered continuous post-launch support and iterative enhancements, ensuring high availability and alignment with evolving business needs.

INTERNSHIPS

Software Testing Engineer, SPRobotics, Chennai, India

Nov 2019 – Dec 2019

- Developed an automated testing suite using Selenium and Python for key user workflows, reducing manual testing time and improving test reliability.
- Conducted comprehensive manual testing based on the Software Requirements Specification (SRS), identifying and documenting bugs for the development team to resolve.

PROJECTS

NanoChat Disease Detection System, Arizona State University, Tempe, AZ

Feb 2026 – Apr 2026

- Built a conversational medical chatbot that predicts diseases from natural language symptom descriptions, achieving 83.78% top-1 and 97.3% top-5 accuracy across 721 disease classes.
- Developed a hybrid symptom extraction pipeline with synonym matching and contextual body-part inference to convert free-text input into structured symptom vectors for an XGBoost classifier integrated with a 561M-parameter transformer LLM (nanochat).

Unified Chest X-ray Analysis Pipeline, Arizona State University, Tempe, AZ

Sep 2025 – Nov 2025

- Developed deep learning models for chest X-ray classification, segmentation, and lesion localization using Swin Transformer, Swin-UNet, and Mask R-CNN.
- Trained and evaluated on three public medical imaging datasets (e.g., ChestX-ray14, SIIM-ACR, VinDr-CXR) with performance comparable to published baselines.
- Built an end-to-end pipeline including data preprocessing, augmentation, model training, and visualization of predictions.

Web Application for Disease Detection, Project Lead, SRM Easwari Engineering College, Chennai, India

Jan 2021 – Apr 2021

- Built a web application for detecting Brain Tumor and Tuberculosis from Brain MRI scan and Chest X-ray.
- Trained ResNet, a deep learning classifier model, for detecting brain tumors. Trained MobileNetV2, another deep learning classifier model, for tuberculosis detection.
- Deployed the models in a web application created using HTML, CSS for frontend and Javascript for backend.
- Led the team, built a robust system architecture allowing for future expansion for detection of additional diseases, managed back-end.

Prediction of Cardiovascular Disease using ML, Project Lead, SRM Easwari Engineering College, Chennai, India

Feb 2020 – Apr 2020

- Used several machine learning classification algorithms like Logistic Regression, Naive Bayes, Decision Tree, SVM, Random Forest to predict cardiovascular disease and compare their accuracies. As a project lead, managed the backend and guided my team members.

CONFERENCE AND PUBLICATIONS

- Published a journal "Online Aid for Detecting Brain Tumor and Tuberculosis using Deep Learning" in IEEE Xplore. (Feb 2022)
- Participated in the 2nd international e-conference on Information, Communication, and Networking [eICIN - 2021]. (Apr 2021)
- Presented in the international conference on Forensics, Analytics, Big Data & Security (FABS - 2021). (Dec 2021)